

## OpenBlue Platform — Johnson Controls

**Dataset:** FullRawChillerPlantSystem\_SAMPLE.xlsx

**Scope:** Dataset Structure · Data Quality · KPI Recommendations

**Date:** April 2026

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Individual Assignment — IoT Sensor Data Analysis

# 1. Dataset Structure & Contents

The repository contains three Excel files that together describe the chilled water plant's operational data and sensor registry.

## 1.1 FullRawChillerPlantSystem\_SAMPLE.xlsx — Time-Series Sensor Data

|                                              |                                            |                                                         |
|----------------------------------------------|--------------------------------------------|---------------------------------------------------------|
| <div>46,751</div> <div>Total data rows</div> | <div>20</div> <div>Columns (signals)</div> | <div>~5.5 mo</div> <div>Date range (Jun–Nov 2011)</div> |
|----------------------------------------------|--------------------------------------------|---------------------------------------------------------|

Records are sampled at a nominal **3-minute cadence** from **2011-06-09 14:54** to **2011-11-26 12:18**. Each row captures a snapshot of the entire chiller plant at that instant.

| Column Group         | Signal Names                                                       | Physical Meaning                                                |
|----------------------|--------------------------------------------------------------------|-----------------------------------------------------------------|
| Condenser Water Loop | CondenserWaterReturnTemp<br>CondenserWaterSupplyTemp               | Water temperatures entering and leaving the cooling towers      |
| Chiller Plant System | CP_ChillerPlantSystem_Efficiency<br>CP_ChillerPlantSystem_kw       | Overall plant efficiency (COP/ kW per ton) and total power draw |
| Chilled Water Pumps  | CP_TotalChilledWaterPump_Efficiency<br>CP_TotalChilledWaterPump_kw | Pump efficiency and power serving building cooling coils        |
| Chillers (Total)     | CP_TotalChiller_Efficiency<br>CP_TotalChiller_kw                   | Aggregate chiller                                               |

| Column Group          | Signal Names                                                                              | Physical Meaning                                       |
|-----------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------|
|                       |                                                                                           | performance across all chiller units                   |
| Condenser Water Pumps | <div>CP_TotalCondenserWaterPump_Efficiency</div> <div>CP_TotalCondenserWaterPump_kw</div> | Pump power circulating water through cooling towers    |
| Cooling Tower System  | <div>CP_TotalCoolingTowerSystem_Efficiency</div> <div>CP_TotalCoolingTowerSystem_kw</div> | Fan/spray system rejecting heat to atmosphere          |
| Weather Station 1 & 2 | Humidity, Wet-bulb Temp, Dry-bulb Temp (×2)                                               | Redundant outdoor weather sensors for cross-validation |

## 1.2 uwc\_meta.xlsx — Building Sensor Registry

Contains **988 sensor point definitions** for the broader building system (not just the chiller plant). Each row maps a raw BACnet UID to human-readable names, location, equipment tag, and data class.

| Category             | Count | Location Breakdown (top)    |
|----------------------|-------|-----------------------------|
| VAV Zone Airflow     | 343   | BB: 291 · BD: 212 · BC: 122 |
| VAV Zone Temperature | 342   | BA: 115 · CP: 73 · IB: 69   |
| Airside              | 230   | BH: 64 · SC: 42             |
| Chiller Plant        | 54    |                             |
| Chiller Plant System | 19    |                             |

## 1.3 uwc\_pm\_meta.xlsx — Power Meter Registry

Lists **78 power meter points** covering the building's electrical submeter hierarchy.

| Meter Type                        | Count |
|-----------------------------------|-------|
| AHU Submeters                     | 36    |
| Lighting Submeters                | 22    |
| Block (Main) Meters               | 10    |
| Section Submeters                 | 6     |
| "LightingSubmeter" (typo variant) | 4     |

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## 2. Data Quality Issues, Missing Data & Anomalies

### 2.1 Schema / Structural Problems

**Missing column header for timestamp.** The first column (the datetime index) has `None` as its header. Any automated script or pipeline that reads this file by column name will silently skip or misprocess the timestamps.

- **Ghost empty column:** Column index 19 is labelled `Unnamed: 0` and contains no data in any row — a pandas export artifact. It adds noise and may confuse tools that infer column count from headers.
- **No units documented:** It is never stated in the file whether temperatures are °C or °F, or whether efficiency is expressed as COP, kW/ton, or another metric. (Context strongly suggests °C and a dimensionless COP ratio capped at 2.0, but this must be assumed by the analyst.)

### 2.2 Zeros Used as a "No Data" Sentinel

**14.5% of all rows (6,781 rows)** have every sensor value equal to exactly 0. These are not valid physical readings — 0°C condenser water is implausible for a tropical/summer climate — yet the file uses 0 as a proxy for "system offline" or "sensor not reporting." This makes it impossible to distinguish a legitimate zero from a missing value without domain knowledge.

| Column                        | Zero Count | NULL Count | Valid Readings | Max Value |
|-------------------------------|------------|------------|----------------|-----------|
| CondenserWaterReturnTemp      | 6,787      | 0          | 39,964         | 46.4°C    |
| CondenserWaterSupplyTemp      | 6,787      | 0          | 39,964         | 41.7°C    |
| ChillerPlantSystem_Efficiency | 16,484     | 3          | 30,264         | 2.0       |
| ChillerPlantSystem_kW         | 13,272     | 1          | 33,478         | 947 kW    |
| TotalChilledWaterPump_kW      | 13,291     | 1          | 33,459         | 37.0 kW   |

| Column                      | Zero Count | NULL Count | Valid Readings | Max Value |
|-----------------------------|------------|------------|----------------|-----------|
| TotalChiller_kW             | 16,494     | 1          | 30,256         | 823.5 kW  |
| TotalCondenserWaterPump_kW  | 14,136     | 2          | 32,613         | 34.6 kW   |
| TotalCoolingTowerSystem_kW  | 16,118     | 1          | 30,632         | 61.0 kW   |
| WeatherStation1_OutdoorTemp | 7,061      | 3          | 39,687         | 32.6°C    |
| WeatherStation2_OutdoorTemp | 7,061      | 3          | 39,687         | 32.9°C    |

Note also the **first 13 rows** (2011-06-09 14:54–15:30) are all-zero — logging started before the system was operational.

## 2.3 Frozen / Stale Sensor Data

The `CondenserWaterReturnTemp` column reports the **identical value for 3,283 consecutive rows** — approximately **6.8 continuous days** — starting 2011-06-17 16:51. A real sensor reading physically cannot stay perfectly constant for that duration. This is a data logger or communication freeze, not a valid steady state. The same freeze affects all other columns simultaneously (they all show identical all-zero blocks), confirming a logging outage rather than a coincidental operational plateau.

## 2.4 Timestamp Gaps (Missing Records)

The nominal 3-minute cadence is broken in **35 places**. Selected gaps:

| Gap Start        | Gap End          | Missing Duration               |
|------------------|------------------|--------------------------------|
| 2011-07-17 08:48 | 2011-07-18 08:42 | <b>1,434 min (≈1 full day)</b> |
| 2011-07-19 17:57 | 2011-07-20 10:09 | <b>972 min (≈16 hours)</b>     |
| 2011-07-18 13:15 | 2011-07-18 15:33 | 138 min                        |
| 2011-07-20 10:12 | 2011-07-20 11:24 | 72 min                         |
| 2011-07-19 14:48 | 2011-07-19 15:42 | 54 min                         |
| 2011-07-04 14:09 | 2011-07-04 14:30 | 21 min                         |

Any naive cumulative energy calculation or rolling-average KPI that does not account for these gaps will produce incorrect results.

## 2.5 Efficiency Values Hard-Capped at 2.0

All five efficiency columns share a hard ceiling of exactly **2.0**. During high-performance operating periods, the true COP may exceed this value but is simply not recorded. The cap affects 33–262 readings per channel (up to 0.9% of valid readings), making peak-efficiency analysis unreliable.

## 2.6 Physical Consistency Flag

Some rows show `TotalChiller_kw = 0` while `TotalCondenserWaterPump_kw > 0`. Running condenser pumps with no chiller load is unusual and likely represents startup/shutdown transition states, but the data provides no mode flag to confirm this interpretation.

## 2.7 Power Meter Metadata Typo

In `uwc_pm_meta.xlsx`, the `PointType` field contains both `"LightingSubmeter"` (22 records) and `"LIghtingSubmeter"` (4 records — capital I). Any GROUP BY or filter on this field will silently miss 4 meters, under-reporting lighting energy by roughly 15% of that submeter category.

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# 3. Recommended KPIs for JCI OpenBlue

## 3.1 Energy Efficiency KPIs

| KPI                                       | Definition                                                           | Source Signals                                                                  | Why It Matters                                                           |
|-------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>Plant-Level COP</b>                    | Total cooling output (tons) ÷ total electrical input (kW)            | <code>CP_ChillerPlantSystem_kW</code> + chilled water BTU meter                 | Single most important efficiency metric; benchmarks against ASHRAE 90.1  |
| <b>Chiller-Only COP</b>                   | Chiller cooling output ÷ chiller kW                                  | <code>CP_TotalChiller_Efficiency</code> , <code>CP_TotalChiller_kW</code>       | Isolates compressor performance from auxiliary loads for fault diagnosis |
| <b>Integrated Part-Load Value (IPLV)</b>  | Weighted-average efficiency across defined load bins (25/50/75/100%) | All efficiency + kW columns                                                     | AHRI 550/590 standard; better real-world metric than rated full-load COP |
| <b>Cooling Tower Approach Temperature</b> | Condenser supply temp – outdoor wet-bulb temp                        | <code>CondenserWaterSupplyTemp</code> , <code>WeatherStation_WetbulbTemp</code> | Measures tower fouling and fan degradation; tighter approach =           |



| KPI                                          | Definition                                          | Source Signals                                         | Why It Matters                                                        |
|----------------------------------------------|-----------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------------|
|                                              |                                                     |                                                        | better efficiency                                                     |
| <b>Condenser Water <math>\Delta T</math></b> | Return temp – supply temp across the condenser loop | CondenserWaterReturnTemp ,<br>CondenserWaterSupplyTemp | Low $\Delta T$ syndrome indicates poor flow distribution or bypassing |
| <b>Specific Fan Power (SFP)</b>              | Cooling tower kW ÷ condenser water flow rate        | CP_TotalCoolingTowerSystem_kw                          | Benchmarks cooling tower fan efficiency independent of load           |

### 3.2 Sustainability KPIs

| KPI                                               | Definition                                             | Why It Matters                                                                     |
|---------------------------------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>kWh per Cooling Ton-Hour</b>                   | Cumulative plant energy ÷ cumulative cooling delivered | Normalised energy intensity; tracks improvement over time or across sites          |
| <b>Carbon Intensity (kgCO<sub>2</sub>/ton-hr)</b> | kWh × local grid emission factor                       | Direct input to ESG reporting and net-zero commitments                             |
| <b>Peak Demand Reduction %</b>                    | (Baseline peak kW – actual peak kW) ÷ baseline peak kW | Quantifies demand-response and load-shifting value; reduces utility demand charges |
| <b>Water Consumption Index</b>                    | Cooling tower make-up water ÷ ton-hours of cooling     | Water stewardship metric increasingly required in sustainability disclosures       |

### 3.3 Equipment Reliability KPIs

| KPI                                       | Definition                                                                               | Why It Matters                                                                                   |
|-------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Sensor Data Availability %</b>         | Valid (non-zero, non-stale, non-NULL) intervals ÷ total expected intervals               | Quantifies how much of the data is actually usable; a precondition for all other KPIs            |
| <b>Equipment Uptime %</b>                 | Hours with chiller/pump kW above minimum threshold ÷ total scheduled hours               | Traditional reliability metric; baseline for maintenance scheduling                              |
| <b>Stale-Data Event Count</b>             | Number of incidents where any sensor reports the same value for >N consecutive intervals | Flags data logger freezes and sensor communication dropouts before they corrupt analyses         |
| <b>Timestamp Gap Frequency</b>            | Count and total duration of missing 3-minute intervals per month                         | Measures logging reliability; gaps invalidate cumulative energy totals                           |
| <b>Efficiency Ceiling Exceedance Rate</b> | % of intervals where efficiency = exactly 2.0 (the sensor cap)                           | Flags how often true performance is hidden by the reporting ceiling; prompts sensor range review |
| <b>Chiller Load Factor</b>                | Average chiller kW ÷ rated chiller capacity kW                                           | Identifies chronic under-loading (degraded COP) or over-loading (reliability risk)               |

## 4. Key Findings Summary

**What AI tools typically get right:** Column identification, basic statistics, spotting NULL values, and generating generic KPI lists for chiller plants.

**What AI tools typically miss:**

- Zeros-as-sentinel vs. true zeros (requires domain knowledge to distinguish)
- The 6.8-day frozen sensor run (requires sequence analysis, not just column stats)

- The missing timestamp header (only visible by inspecting raw cell A1)
- The efficiency hard cap at 2.0 (appears as a valid max value without context)
- The metadata type `LightingSubmeter` (easy to overlook without a GROUP BY)
- Physical inconsistency between chiller kW = 0 and condenser pump kW > 0

Generated from direct programmatic inspection of FullRawChillerPlantSystem\_SAMPLE.xlsx, uwc\_meta.xlsx, and uwc\_pm\_meta.xlsx · April 2026